

TASK 1: Analysis of Bitcoin Fear & Greed Index

Methodology:

Time Series Engineering: Standardized historical data was used, with temporal feature engineering (month, year, day) for potential cyclical patterns. Rolling moving averages for 7 days and 30 days were also computed for momentum and day-to-day price volatility.

Statistical Modeling: A Markov-like model was developed for day-over-day probabilistic persistence of emotions, defined as the probability of Fear given Fear.

Data Visualization: Multi-layered visual representations were developed, including distribution histograms, heat maps, and streak density plots, for visual delineation of market regimes.

Insights:

Sentiment has a very high autocorrelation, where the market enters a Fear or Greed regime, and there is a very high probability that the market will stay in that regime the following day.

Fear is the baseline state, where historically the market spends a lot more time in either Fear or Neutral compared to Greed.

From the data for 2023 and 2024, a very strong bias towards Greed is evident, which is consistent with the overall bullish macro-economic environment.

Extreme values, i.e., values < 25 or > 75 , are very infrequent.

Strategy Recommendations:

Momentum-based trading within regimes: Due to the autocorrelation of the market, trading strategies should align with the moving average crossovers for 7 days and 30 days, rather than going against the market sentiment.

Mean-Reversion Strategies: Counter-trend algorithms should be used only when the market is in the infrequent zones of either Fear < 25 or Greed > 75 , where historically the market has shown exhaustion and is poised for a reversal.